

Literature Review

SentinelLite: A Lightweight SIEM with Rule-Based Detection and ML Anomaly Scoring

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Background

NIST SP 800-92 emphasizes that log management is a practical security capability covering generation, transmission, storage, analysis, and disposal of log data [1]. SentinelLite narrows this broad discipline into an academic SIEM that focuses on parsing, correlation, triage, and reporting.

Modern SIEM work increasingly combines deterministic rules with anomaly detection. Deterministic rules remain useful for known attack patterns, while anomaly detection helps surface behavior that differs from a local baseline.

For a semester project, the main design challenge is not only detection accuracy; it is also operational completeness. A useful SIEM prototype should show how evidence moves from log source to parser, from parser to alert, and from alert to analyst action.

Machine Learning for Log Anomaly Detection

Isolation Forest is suitable for a lightweight project because it isolates unusual observations rather than requiring a large labeled attack dataset [2]. This aligns with student lab conditions where large labeled enterprise logs are rarely available.

Recent log-analysis systems such as LogLead highlight the importance of loading, enhancing, and evaluating logs through reproducible pipelines [3]. SentinelLite follows the same broad pipeline at a smaller scale: load logs, normalize fields, extract features, detect alerts, and evaluate against a labeled sample.

Recent work on log anomaly detection shows that advanced methods can improve benchmark performance, but practical adoption still depends on explainability, usability, and evaluation design [10], [11].

Deep sequence models and transformer-based approaches can learn richer temporal context, but they require more training data, more compute, and more explanation effort. SentinelLite intentionally uses a classical unsupervised method so the complete system remains feasible on a lab machine.

Threat Mapping and Rule-Based Detection

The MITRE ATT&CK framework provides a vocabulary for mapping detections to adversary behavior. SentinelLite maps brute-force login bursts to T1110 [4], port-scan patterns to T1046 [5], web exploitation probes to T1190 [6], suspicious PowerShell to T1059.001 [7], and log-clearing behavior to indicator-removal activity [8].

Rule-based detection provides precise explanations for known conditions. In SentinelLite, rules produce evidence strings, recommended responses, severity, confidence, and attack phase so the analyst can understand why an alert exists.

A rule-only system can miss unknown behavior. An ML-only system can produce alerts that are hard to defend. The chosen hybrid approach is therefore a practical compromise: rules provide high interpretability, while anomaly scoring provides broader coverage.

Evaluation Practices

Security testing should report both what was detected and what was missed. The brief explicitly rewards structured test plans, tabulated results, and false-positive/false-negative discussion. SentinelLite therefore includes expected alert labels for the safe sample dataset and computes group-level precision, recall, and F1 score.

The evaluation is intentionally described as sample-set evaluation, not as a universal production claim. This avoids overstating accuracy while still demonstrating that the system is testable and repeatable.

Research Gap

Full SIEM platforms are powerful but heavy for small labs. Research prototypes may focus on algorithms but not on a complete student-demo workflow. SentinelLite fills this gap by combining modest but complete components: multi-source parsing, rules, ML anomaly scoring, SQLite persistence, UI, email alerts, PDF reporting, and evaluation.

The project gap is therefore implementation-oriented: a lightweight, explainable, low-cost SIEM workflow that can be demonstrated live, defended technically, and extended after the semester.

Comparison Table

Approach	Strength	Limitation	SentinelLite Decision
Traditional log management [1]	Strong operational guidance	Does not prescribe a student implementation	Use simple tools as foundation.
Isolation Forest [2]	Unsupervised and efficient	Can flag benign unusual behavior	Use with explainable rules.
Benchmark systems [3]	Reproducible evaluation	More complex than needed	Adopt separate concepts only.
Deep log models [10], [11]	Strong representation learning	Higher compute/data requirements	Leave as future work.

References

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